

AI News Agent

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Objective

AI moves faster than any single reader can track — new models, papers, and tools surface daily across scattered sources, with no dedicated feed and no visible link between a new release and what came before it. The AI News Agent ingests AI-specific news, research papers, and tool releases, and turns them into short, original summaries — each linked to its source, to related stories, and to the underlying paper where one exists.

Its core objective is to build a **Knowledge Graph of AI** alongside the feed: foundational topics are seeded from standard AI literature and courses, and new nodes are inserted dynamically as fresh news and papers surface new topics — mapping the problem we're solving directly onto the graph as it grows. Edges capture how topics precede, build on, or vary from one another, so the graph doubles as a traversable chronology of the field. This is also meant to improve discovery of broader AI applications: a user following one topic can traverse chronologically into an adjacent, less-covered frontier — for example from Computer Vision into Physical AI.

1 Data Description

DATA SOURCE	FORMAT	PROCESSING
AI news & announcements NewsAPI / GNews, RSS from AI-focused outlets, GDELT	JSON / RSS	Filtered for AI relevance, deduplicated across outlets, tagged by AI subfield
Research papers arXiv API (cs.AI, cs.CL, cs.CV, cs.RO), Semantic Scholar API	JSON + abstract / PDF text	Embedded and matched to related news by semantic similarity; chunked for retrieval
Tool & framework releases GitHub API (trending repos, releases), MCP server registries	JSON	Parsed for "what changed," mapped to the nearest knowledge-graph node

PIPELINE

Ingest (scheduled pollers, hourly/daily) → clean & dedupe → LLM classify & tag by AI subfield and entities → embed for similarity linking → summarize in original phrasing → store (relational DB for items, vector store for embeddings, graph DB for topic relationships) and link into the knowledge graph via nearest-node matching.

2 Architecture

Four layers: scheduled ingestion, a processing/enrichment pipeline, an API backend, and the web frontend.

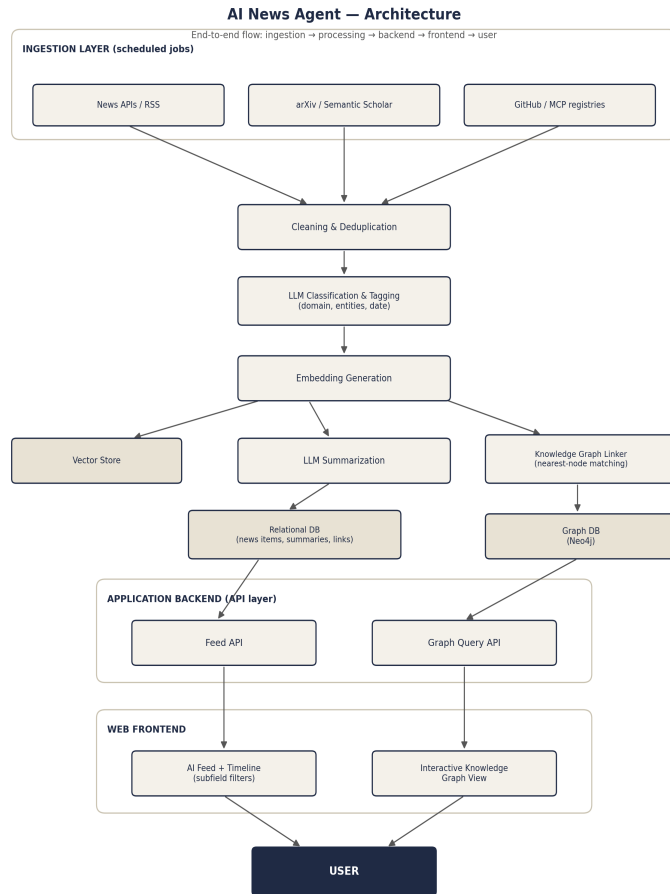


FIG. 1 — end-to-end flow from ingestion through to the feed and knowledge-graph explorer

Flow explanation

- **Ingestion Layer** continuously pulls raw data from news, research, and code/tool sources.
- **Processing pipeline** cleans, classifies, embeds, summarizes, and links each item — both to related news (via vector similarity) and to the knowledge graph (via nearest-node matching).
- **Backend** exposes two main services: the personalized feed and a graph query service for the visualization.
- **Frontend** renders the domain/timeline-filtered feed and a separate interactive graph explorer view.

3 Tech Stack

LAYER	TECHNOLOGY
Backend & orchestration	Python, FastAPI, LangChain / LangGraph (Deep Agents pattern)
LLM provider	Claude API (Anthropic)
Storage	PostgreSQL (relational), ChromaDB / Pinecone (vector), Neo4j (knowledge graph)
Scheduling	Airflow / cron-based workers for periodic ingestion
Frontend	React.js, Tailwind CSS, D3.js / react-force-graph for the graph view
Data sources	NewsAPI, GNews, GDELT, arXiv API, Semantic Scholar API, GitHub API
Deployment	Docker, deployed on AWS / GCP / Render